

Algebra 1 – Unit 3 Test B

Name: _____

Solving Equations

1. Solve for x .

$$x + 6 = 20$$

2. Solve for y .

$$4y - 7 = 17$$

3. **Word Problem:**

After paying a \$5 service fee, the total for four identical sketchbooks was \$29.

Write and solve an equation to find the price of each sketchbook.

Multi-Step Equations

4. Solve for x .

$$3(x - 2) = 12$$

5. Solve for p .

$$2(p + 6) - 4 = 14$$

6. Solve for r .

$$6r + 5 = 3r + 17$$

7. Solve for m .

$$3(m - 2) = 2m + 8$$

8. Solve for t .

$$5(t + 2) - 8 = 5t - 3$$

Solving Inequalities

9. Solve and graph the solution on a number line.

$$3x + 2 > 11$$

10. Solve and graph the solution on a number line.

$$-2z \geq 6$$

Literal Equations

11. Solve for h .

$$V = \frac{1}{3}\pi r^2 h$$

12. Solve for t .

$$C = 4h + 2t$$

13. Solve for x .

$$y = m(x - b)$$